

CURRICULUM VITAE

PERSONAL INFORMATION

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GitHub	https://github.com/lysandre995

WORK EXPERIENCE

2024-04 - Present

Position held: Application Engineer

Main activities and responsibilities:

- Design and development of software applications and integration layers for automated systems, including architecture definition, communication protocols, and data models
- Contribution to the evolution of the company's application framework (.NET/WPF), focusing on modular architecture, extensibility, and web technology integration
- R&D activities: evaluation and adoption of new data storage technologies, CI/CD and containerization practices
- Management and evolution of company's private git infrastructure, introduction to private Package/Container Registry and advanced Git automatic workflows

- Internal reference for Linux environments, web technologies, and AI-assisted development workflows

Focus: Reference point for application software, R&D, and development infrastructure

Name and address of employer: Microtecnica Engineering, Novazzano, Switzerland

Type of business or sector: Swiss engineering company specializing in high-tech industrial solutions, robotics, and automated systems

Technologies used: Git, GitLab CI/CD, Docker, InfluxDB, Linux, gRPC, MQTT, Redis, .NET, WPF

2023-10 - 2024-04

Position held: Software Architect

Main activities and responsibilities:

- Led the complete redesign and rewrite of a distributed system for electricity consumption/production data collection and processing
- Coordinated development team and defined end-to-end system architecture
- Designed meter acquisition layer using Modbus protocol and IoT router configuration
- Architected on-premise cloud infrastructure with VMs and Docker Swarm
- Developed data visualization platform using Angular and NestJS
- Implemented automated billing data generation and OCPP server for EV charging stations

Focus: Architecture redesign of the data platform, from meter acquisition to billing

Name and address of employer: YouPower AG, Mezzovico, Switzerland

Type of business or sector: Swiss energy technology company focused on smart metering and energy community solutions

Technologies used: Angular, NestJS, TypeScript, Docker Swarm, MongoDB, Redis, Modbus, RouterOS, WireGuard, OCPP

2022-02 - 2023-09

Position held: Full Stack Developer

Main activities and responsibilities:

- Contributed to Leapp, an open-source application for secure cloud access management in DevOps environments
- Designed and implemented AWS-based solutions using TypeScript, including e-commerce-to-ERP integration flows
- Developed microservices-oriented document processing system (Spring Boot) for Italian electronic invoicing standards

Focus: Open-source work on Leapp and cloud integrations for enterprise clients

Name and address of employer: Forge Srl, Varese, Italy

Type of business or sector: Italian software house specializing in cloud-native solutions and enterprise integrations

Technologies used: TypeScript, AWS, Spring Boot, Java, Node.js

2021-05 - 2022-01

Position held: Software Developer

Main activities and responsibilities:

- Contributed to design and evolution of automation systems for automated warehouses
- Integrated and optimized mission management system for a leading Italian food producer/distributor
- Conducted operational research for technology stack modernization

Focus: Mission management system for automated warehouses

Name and address of employer: Arsinform Srl, Milan, Italy

Type of business or sector: Italian company specializing in industrial automation and warehouse management software

Technologies used: Oracle, PL/SQL

EDUCATION AND TRAINING

2020 - 2026

Title of qualification awarded: B.Sc. Computer Engineering

Principal subjects/occupational skills covered: Computer engineering, software development, system architecture

Name and type of organization providing education and training: Politecnico di Milano, Milan, Italy

2014 - 2020

Title of qualification awarded: Medicine and Surgery (partial)

Principal subjects/occupational skills covered: Medicine and surgery

Name and type of organization providing education and training: University of Milan, Milan, Italy

2009 - 2014

Title of qualification awarded: Classical Studies Diploma (Maturità Classica)

Principal subjects/occupational skills covered: Latin, Ancient Greek, classical literature, philosophy

Name and type of organization providing education and training: Liceo Classico Alessandro Volta, Como, Italy

PERSONAL SKILLS AND COMPETENCES

Mother tongue

Italian

Other languages

Language	Level
English	C1 (Professional)
French	A2 (Elementary)
Latin	Reading
Ancient Greek	Reading

Technical skills and competences

Software Architecture & System Design: Design of modular systems: boundaries, responsibilities, data models, and how a system is meant to change over time.

Key skills: System Design, Modular Architecture, Technical Strategy, Domain Modeling

Complex System Integration: Integration of systems across automation, IoT, cloud services, and enterprise platforms: communication protocols, data flows, and orchestration patterns.

Key skills: System Integration, Protocol Design, Data Orchestration, Heterogeneous Systems

Distributed Systems & Cloud-Native Architectures: Distributed systems, microservices, containerized environments, and hybrid cloud/on-prem infrastructures.

Key skills: Microservices, Containerization, Cloud-Native, Hybrid Infrastructure

DevOps & Engineering Practices: Git workflows, CI/CD pipelines, containerization with Docker, and development infrastructure management.

Key skills: CI/CD, GitOps, Containerization, Infrastructure as Code

Applied Research & Technology Evaluation: Evaluation and adoption of new technologies, integration patterns, and AI-assisted workflows.

Key skills: R&D, Technology Evaluation, Innovation, Continuous Improvement

Computer skills and competences

Programming Languages: TypeScript, JavaScript, Java, C#, Lua, Python, Rust, SQL

Frameworks: .NET, NestJS, Node.js, Spring Boot, Angular, Blazor, Vue

Infrastructure: Docker, Docker Swarm, Docker Compose, Kubernetes, GitLab CI/CD, GitHub Actions, AWS

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis, InfluxDB

Architecture: System Architecture, Microservices Design, Domain-Driven Design, Event-Driven Architecture, API Design

ADDITIONAL INFORMATION

Courses

- **AWS Solutions Architect – Associate (Training Course)** (beSharp Srl, 2022-11)

Open Source Contributions

- **Leapp** (Noovolari): Feature development during the Forge period (2022-2023)
<https://github.com/Noovolari/leapp>

Projects

- **Alphamon** (Game Framework): Collectible monster-fighting RPG starter kit in Lua/Love2D, with dependency injection, TCP multiplayer, a battle system, and dialog management. MIT-licensed, to be released as open source. *Technologies:* Lua, Love2D
- **Multi-Protocol Integration Framework** (Integration Framework): Framework/application template for multi-protocol integration across distributed, heterogeneous environments. Supports OPC UA, Redis, REST, gRPC, and other protocols for heterogeneous system communication. *Technologies:* .NET, OPC UA, Redis, gRPC, REST
- **AML (Alessandro's Music Language)** (Creative Tool): Text-based music notation system for chiptune composition, with a Game Boy Color-style synthesis framework built on ChuckK and Python CLI utilities. *Technologies:* ChuckK, Custom DSL

University projects:

- **GathorApp** (Web Application): Outing and event management system with geolocation, JWT authentication, real-time chat, and vouchers.
<https://github.com/lysandre995/gathorapp>
- **Green Wheels** (Web Application): Car pooling application, split into frontend, backend, and shared-core repositories. <https://github.com/lysandre995/green-wheels>

Other public projects:

- **Lorenzo Orsenigo Website** (Website): Website for percussionist Lorenzo Orsenigo.
<https://github.com/lysandre995/lorenzo-orsenigo-website>
- **Personal Portfolio** (Website): Personal website and CV platform with JSON-based data backend. <https://alessandroalfano.com>
- **Waterfall Enterprise** (Website): Site of the fictional company that appears in the theatre production Waterfall, written and directed by Ilaria Casati. <https://waterfall-enterprise.org>

Theatre

- **Communication with Teatro Sociale di Como** - Compagnia Teatrale Trebisonda (2026 - Present): Contact person for the company's relationship with Teatro Sociale di Como. Current work covers workshops and "Theatron - una sfida a teatro", part of the Stravolta campaign of the 2026/2027 season.

- **Waterfall, by Ilaria Casati** - Compagnia Teatrale Trebisonda (2025 - 2026): Production staged in four performances between 2025 and 2026. The site of the fictional company featured in the play is listed under Projects.
- **Workshop activities** - Mumble Teatro (2021 - 2025): Workshop cycles with the association, each ending in a final production; four in total.

Interests and Activities

Creative:

- Music: Singing and playing in a rock band; chiptune composition; music notation systems
- Game Development: Indie game development in Lua/Love2D

Physical:

- Sailing
- Soccer Coaching: Youth team coaching

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